

SL2DT-05: Monitor & Alert Conversion

Series: SL2DT — Sumo Logic to Dynatrace | **Notebook:** 5 of 11 | **Created:** April 2026 | **Last Updated:** 10/06/2026

Overview

Goal of this step: rebuild Sumo Monitors in Dynatrace, choosing the right target for each (Anomaly Detection, Workflow with DQL threshold, or Metric Event). This is the notebook where the biggest fidelity-vs-better-approach decisions happen.

The anti-pattern to avoid: 1:1 static-threshold lift-and-shift. It produces noisy alerts, misses real anomalies, and under-uses Dynatrace's capabilities. The pattern to prefer: Anomaly Detection for metric-backed alerts, Workflows for complex multi-condition logic, and Metric Events for simple static thresholds that genuinely are static.

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Prerequisites

Requirement	Details
Audience	Core migration engineers + app-team reviewers
Inputs	<code>inventory/monitors.json</code> , translations from SL2DT-04
Dynatrace access	Platform Token with <code>settings:objects:write</code> , <code>automation:workflows:write</code> , <code>davis:anomaly-detectors:write</code>
Prior reading	SL2DT-04 for query translations; OPLOGS-09 for log-based alert patterns

1. What You'll Produce

Artifact	Purpose
<code>monitors/decision-matrix.csv</code>	Per-monitor: Dynatrace Intelligence vs Workflow vs Metric Event decision
<code>monitors/configs/</code>	Terraform + JSON for Dynatrace Intelligence detectors, Workflows, Metric Events
<code>notifications/action-map.md</code>	Sumo webhook/email → DT notification target map
<code>monitor-rebuild-report.md</code>	Progress per team

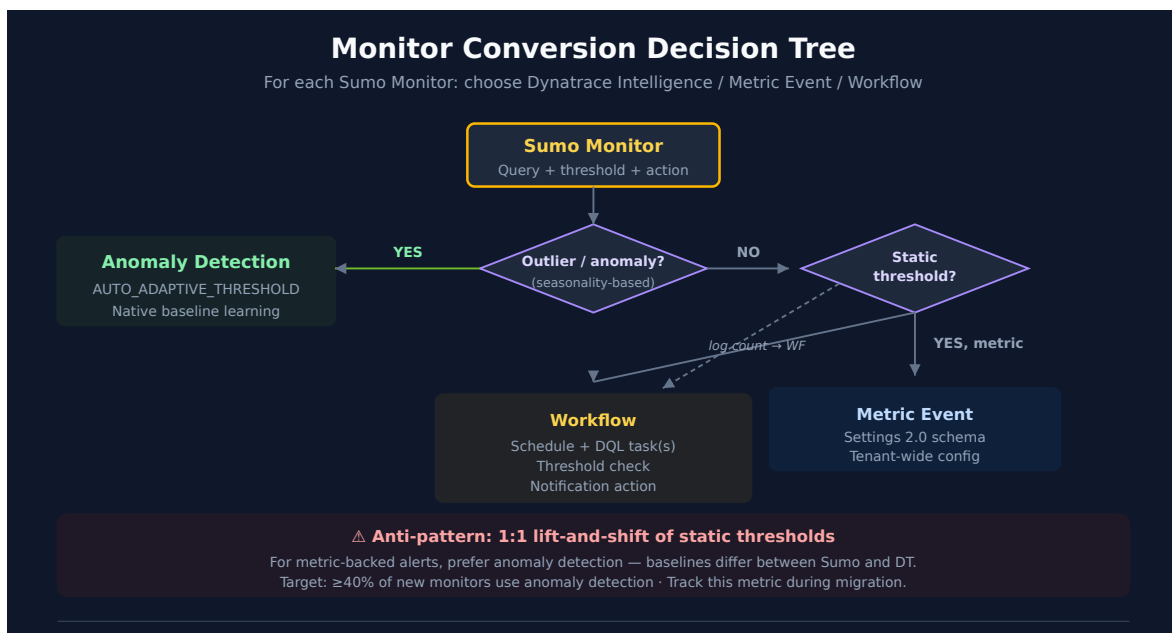
2. The Monitor Conversion Decision Framework

For each Sumo monitor, pick one of three targets:

Decision Rules

Sumo Monitor Type	Target	Why
Outlier / anomaly / seasonality-based	anomaly detection	Native baseline learning, no tuning

Sumo Monitor Type	Target	Why
Static threshold on metric (CPU > 90%)	Metric Event	Lightweight, tenant-wide config
Static threshold on log count	Workflow with DQL	Query runs on schedule, threshold in task condition
Missing data alert	Workflow	Use <code>isNull</code> / <code>count == 0</code> checks
Multi-condition (X AND Y but not Z)	Workflow	Multiple DQL tasks + branching
SLO-backed alert	SLO burn-rate alert (if Dynatrace SLO)	Native SLO product



3. Anomaly Detection — When & How

Dynatrace Intelligence learns the baseline of a metric and alerts when values deviate.
Best for:

- Any Sumo outlier monitor (`outlier _count window=N threshold=K`)
- Static thresholds where the "normal" value drifts seasonally (weekly cycle, business hours)
- Static thresholds that fire too often in Sumo (false positives from workload swings)

Configuration

anomaly detection is configured via the Settings API (`builtin:anomaly-detection.metric-events`) or via the UI. This is the **metric-event** schema, which evaluates Metrics Classic keys. An adaptive or seasonal model needs a `METRIC_SELECTOR` query definition — *"Metric key based query definitions only support static thresholds"* — and every model needs its full set of sample counts plus an event template. For programmatic setup:

```
{
  "schemaId": "builtin:anomaly-detection.metric-events",
  "scope": "environment",
  "value": {
    "enabled": true,
    "summary": "API error rate anomaly",
    "queryDefinition": {
      "type": "METRIC_SELECTOR",
      "metricSelector":
"custom.metric.api_error_rate:splitBy(\"dt.entity.host\"):avg"
    },
    "modelProperties": {
      "type": "AUTO_ADAPTIVE_THRESHOLD",
      "signalFluctuation": 1.0,
      "tolerance": 4.0,
      "alertCondition": "ABOVE",
      "alertOnNoData": false,
      "samples": 5,
      "violatingSamples": 3,
      "dealertingSamples": 5
    },
    "eventTemplate": {
      "title": "API error rate anomaly",
      "description": "The API error rate is above its learned baseline.",
      "eventType": "ERROR",
      "davisMerge": true,
      "metadata": []
    }
  }
}
```

Sources: [builtin:anomaly-detection.metric-events](#) schema (DT docs).

Sumo outlier → anomaly detection example

Sumo:

```
_sourceCategory=prod/api | timeslice 1m | count | outlier _count window=10  
threshold=3
```

Dynatrace Intelligence:

1. Create a custom metric from the DQL equivalent (via metric extraction in OpenPipeline, or via `makeTimeseries` into an extracted metric).
2. Configure anomaly detection on that metric with `AUTO_ADAPTIVE_THRESHOLD` mode.
3. Attach the Sumo monitor's notification channel to the detected problem.

Metric Extraction from Logs

If the Sumo monitor was log-count-based, extract a metric first:

```
{  
  "schemaId": "builtin:logmonitoring.log-custom-metric",  
  "scope": "environment",  
  "value": {  
    "key": "custom.metric.api_error_count",  
    "enabled": true,  
    "query": "sumo.source_category = \"prod/api\" AND status = \"ERROR\"",  
    "measure": {"type": "OCCURRENCE"},  
    "dimensions": ["dt.entity.host"]  
  }  
}
```

Then apply anomaly detection to the extracted metric.

Validation

```
// Confirm metric extraction is producing data  
timeseries c = avg(custom.metric.api_error_count), from:-1h, by:  
{dt.entity.host}  
| fieldsAdd c_max = arrayMax(c)  
| filter c_max > 0
```

4. Metric Events — Simple Static Thresholds

When the threshold truly is static and the underlying metric is well-defined:

- CPU > 90%
- Memory available < 5%
- Disk full
- Certificate expiry < 30 days

Use Metric Events via `builtin:anomaly-detection.metric-events` :

```

{
  "schemaId": "builtin:anomaly-detection.metric-events",
  "scope": "environment",
  "value": {
    "enabled": true,
    "summary": "Host CPU > 90%",
    "queryDefinition": {
      "type": "METRIC_KEY",
      "metricKey": "builtin:host.cpu.usage",
      "aggregation": "AVG",
      "entityFilter": {
        "dimensionKey": "dt.entity.host",
        "conditions": [
          {"type": "HOST_GROUP_NAME", "operator": "EQUALS", "value": "prod-
web"}
        ]
      }
    },
    "modelProperties": {
      "type": "STATIC_THRESHOLD",
      "threshold": 90.0,
      "alertCondition": "ABOVE",
      "alertOnNoData": false,
      "samples": 5,
      "violatingSamples": 3,
      "dealertingSamples": 5
    },
    "eventTemplate": {
      "title": "Host CPU > 90%",
      "description": "CPU usage has been above 90% for 3 of the last 5
minutes.",
      "eventType": "RESOURCE",
      "davisMerge": true,
      "metadata": []
    }
  }
}

```

The schema's only scope is `environment` ; narrow the alert to particular hosts with the `entityFilter` (here a host group), not with an entity ID as the scope. Metric events read Metrics Classic, so the key is `builtin:host.cpu.usage` — the Grail key `dt.host.cpu.usage` is for DQL.

When NOT to use Metric Events

- Threshold depends on workload (use Dynatrace Intelligence)
- Condition involves multiple metrics (use Workflow)
- Alert requires custom notification payload (use Workflow)

5. Workflows — Complex or Multi-Condition Alerts

Workflows are the equivalent of Sumo's more complex monitors. Use when:

- Alert logic needs multiple DQL queries
- Conditional branches required
- Custom notification payload (ServiceNow incident with specific fields, etc.)
- Scheduled evaluation (every N minutes)

Workflow Structure

The schedule (cron `* /5 * * * *`, every 5 minutes) is the workflow's trigger, set in the editor rather than written into the task list. The HTTP task authenticates with a Credential Vault credential selected in its **Authentication** field.

```

tasks:
  - name: check_error_rate
    action: dynatrace.automations:execute-dql-query
    input:
      query: |
        fetch logs, from:-5m
        | filter sumo.source_category == "prod/api"
        | summarize {total = count(), errors = countIf(contains(content,
"error", caseSensitive:false))}
        | fieldsAdd error_pct = 100.0 * toDouble(errors) / toDouble(total)
  - name: check_threshold
    action: dynatrace.automations:http-function # ServiceNow Table API – see
$6
    predecessors: [check_error_rate]
    conditions:
      states:
        check_error_rate: SUCCESS
      custom: '{{ result("check_error_rate").records[0].error_pct > 5 }}'
    input:
      method: POST
      url: https://&lt;instance&gt;.service-now.com/api/now/table/incident
      headers:
        Content-Type: application/json
      payload: |
        {
          "short_description": "API error rate {{
result('check_error_rate').records[0].error_pct }}%",
          "assignment_group": "Payments Platform",
          "priority": "2"
        }

```

Scheduled Search → Workflow

A Sumo scheduled search (query + schedule + action) maps directly:

Sumo Scheduled Search Field	Workflow Equivalent
Query	Execute DQL Query task (<code>dynatrace.automations:execute-dql-query</code>)
Schedule (run every N)	Schedule trigger with a cron rule, set in the editor
Alert condition	<code>conditions.custom</code> on the notification task (a Jinja expression)

Sumo Scheduled Search Field	Workflow Equivalent
Webhook action	HTTP Request task (<code>dynatrace.automations:http-function</code>) or a connector action
Email action	Send email task (<code>dynatrace.email:send-email</code>)

DQL in Workflow Tasks

The translated DQL from SL2DT-04 goes directly here. Wrap long queries in `|` to preserve formatting:

```
tasks:
- name: find_slow_transactions
  action: dynatrace.automations:execute-dql-query
  input:
    query: |
      fetch logs, from:-5m
      | filter sumo.source_category == "prod/api"
      | parse content, "LD? 'latency=' INT:latency"
      | filter latency > 2000
      | summarize c = count(), by:{http.path}
      | sort c desc
      | limit 20
```

6. Rebuilding Notification Actions

Every Sumo monitor has one or more actions. Map each to a Dynatrace notification target.

Sumo Action	Dynatrace Target	Notes
Email	Send email task (<code>dynatrace.email:send-email</code>)	Same recipient list; at most ten per field
Webhook → ServiceNow	HTTP Request task to the Table API, or ServiceNow Create Incident (<code>dynatrace.servicenow:snow-create-incident</code>)	Rebuild incident payload
Webhook → Slack	Slack Send message task (<code>dynatrace.slack:slack-send-message</code>)	Rebuild message format

Sumo Action	Dynatrace Target	Notes
Webhook → PagerDuty	PagerDuty Send event task (<code>dynatrace.pagerduty:send-event</code>)	Rebuild event payload
Webhook → custom HTTP	HTTP Request task (<code>dynatrace.automations:http-function</code>)	
Mobile push	Workflow + Dynatrace Mobile app	

ServiceNow Integration — Specific Patterns

ServiceNow is the most common target and the highest-risk translation (wrong field mapping → misrouted incidents).

Sumo webhook payload:

```
{
  "incident": {
    "short_description": "{{Name}}: {{Description}}",
    "u_affected_service": "{{ClusterName}}",
    "assignment_group": "{{Team}}",
    "priority": "{{Priority}}"
  }
}
```

Dynatrace Workflow equivalent:

```

- name: create_servicenow_incident
  action: dynatrace.automations:http-function
  input:
    method: POST
    url: https://&lt;instance&gt;.service-now.com/api/now/table/incident
    # Authentication: select a Credential Vault credential (Basic) in the
    task's
    # Authentication field. Never put the credential in a static Authorization
    header.
    headers:
      Content-Type: application/json
    payload: |
      {
        "short_description": "API error rate {{
result('check_error_rate').records[0].error_pct }}%",
        "u_affected_service": "{{ result('check_error_rate').records[0]
['dt.entity.service'] }}",
        "assignment_group": "Payments Platform",
        "priority": "2"
      }

```

The request body goes in `payload`, and dotted field names need bracket access (`records[0]['dt.entity.service']`). The HTTP route keeps custom fields such as `u_affected_service`. The ServiceNow Connector's **Create Incident** action (`dynatrace.servicenow:snow-create-incident`) is the alternative when the standard fields are enough. It takes named inputs (`shortDescription`, `description`, `impact`, `urgency`, `category`, `group`, `correlationId`, ...) rather than a raw body. For the query above, `u_affected_service` stays empty unless the query groups by `dt.entity.service`.

Sources: [HTTP request action \(DT docs\)](#) — "Payload : The payload of the HTTP request.", "We strictly advise against providing any static Authorization header and therefore, leak a secret."; [wftpl_sample_servicenow_incident_man.yaml \(Dynatrace GitHub\)](#) — "action: `dynatrace.servicenow:snow-create-incident`".

Verify with a test incident before flipping production traffic.

7. Handling Rare Alert Classes

Missing Data Alerts

Sumo:

```
_sourceCategory=prod/heartbeat | count  
| where _count == 0
```

Dynatrace Workflow:

```
tasks:  
- name: check_heartbeat  
  action: dynatrace.automations:execute-dql-query  
  input:  
    query: |  
      fetch logs, from:-10m  
      | filter sumo.source_category == "prod/heartbeat"  
      | summarize c = count()  
- name: alert_if_missing  
  action: dynatrace.email:send-email  
  predecessors: [check_heartbeat]  
  conditions:  
    custom: '{{ result("check_heartbeat").records[0].c == 0 }}'  
  input:  
    to: ["oncall@example.com"]  
    cc: []  
    bcc: []  
    subject: "No heartbeat from prod/heartbeat"  
    content: "No heartbeat logs from prod/heartbeat in the last 10 minutes."
```

Change Detection Alerts

Sumo:

```
_sourceCategory=audit | logcompare timeshift=24h
```

Dynatrace: Change detection or two-fetch comparison workflow. See OPLOGS-08.

Multi-Metric Correlation

Sumo: two separate monitors with same target; Dynatrace: one Workflow with two DQL tasks + AND condition.

8. Tuning & Noise Reduction

Expect more alerts in the first week post-cutover — baselines aren't learned yet, and Workflow thresholds may be too aggressive.

First-week discipline

1. Monitor alert volume daily. Dashboard the total count of detected problems + Workflow triggers.
2. For every alert, triage: real issue, false positive, or threshold too tight?
3. Adjust thresholds daily until noise drops below Sumo baseline.

anomaly detection — let it learn

Dynatrace Intelligence needs ~2 weeks of data to build a stable baseline. During week 1–2, expect more sensitivity alerts. Keep them in a separate "tuning" severity class if possible; don't page on-call for them.

Checking alert volume

```
// Alert volume over time – compare Sumo baseline
// dt.davis.problems holds one record per problem; `fetch events | filter
event.kind == "DAVIS_PROBLEM"`
// returns every update of every problem and over-counts many times over.
fetch dt.davis.problems, from:-7d
| makeTimeseries problems = count(), interval:24h, time:event.start
```

Silence during load tests

Workflow-based monitors should respect maintenance windows. Settings 2.0 schema `builtin:maintenance.general` applies tenant-wide; configure before load tests, CI/CD deployments, or controlled outages.

Severity mapping

Sumo Priority	DT Severity	Notification
Critical	ERROR / Dynatrace Intelligence P1	PagerDuty + ServiceNow
High	Dynatrace Intelligence P2	ServiceNow
Warning	Dynatrace Intelligence P3	Email
Info	(no alert)	Logged only

9. Step Exit Criteria

G5 — Monitors Rebuilt

- [] Every Sumo monitor has a decision logged (Dynatrace Intelligence / Workflow / Metric Event / retire)
- [] ≥40% of new monitors use anomaly detection (not static thresholds) — track this metric
- [] Top-10 business-critical monitors validated end-to-end (fires on known-bad condition, doesn't fire on normal)
- [] Notification integrations (ServiceNow, Slack, PagerDuty) tested with live test incidents
- [] First-week alert volume ≤ Sumo baseline
- [] App teams trained on monitor rebuild (train-the-trainer — see SL2DT-01)

Next step: SL2DT-06 — Dashboard Conversion (rebuild dashboards using the translation output).

10. References

Dynatrace alerting and intelligence

- [Anomaly detection \(DT docs\)](#)
- [Workflows \(DT docs\)](#)

- [Notifications and alerting \(DT docs\)](#)
- [Problems app \(DT docs\)](#)
- [Maintenance windows \(DT docs\)](#)

Sumo Logic monitor and alert reference (source)

- [Sumo Logic monitors \(Sumo Logic docs\)](#)

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